Morning

Regular Expression

- Definition
- Design
- Equivalence with FA

Arithmetical Expression

$0, 1+2, 3\times(5-2), (56-7)^2, \ldots\ldots$

- Traditional definition
- Inductive definition
  - Any number is a arithmetical expression
  - If **a** and **b** are arithmetical expressions , then so is $a+b, a-b, a\div b, a\times b, a^n, (a)$.

Building Regular Expressions

BASIS

1. ε is a RegExp, denoting the language $\{\varepsilon\}$.

2. ϕ is a RegExp, denoting the language ϕ.

3. For each a in Σ, a is a RegExp and denotes the language $\{a\}$.

Notice : language is a set of strings.

Building Regular Expressions

INDUCTION

1. If E and F are RegExp, denoting the language $L(E)$ and $L(F)$, then $E+F$, EF and E^* are RegExp that denote the languages $L(E) \cup L(F)$, $L(E)L(F)$ and $(L(E))^*$.

2. If E is a RegExp, then so is (E) .

Example 1 Language for

$$r = (a+b)^*(a+bb)$$

$a \rightarrow \{ a \}$, $b \rightarrow \{ b \}$

$a + b \rightarrow \{ a \} \cup \{ b \} = \{a, b\}$

$bb \rightarrow \{ b \} \{ b \} = \{ bb \}$

$a + bb \rightarrow \{ a \} \cup \{ bb \} = \{ a, bb \}$

$(a + b)^* \rightarrow \{ a, b \}^*$

$(a + b)^* (a + bb) \rightarrow \{ a, b \}^* \{ a, bb \}$

$L(r) = \{ a, bb, aa, abb, ba, bbb, \ldots\ldots \}$

Example 2 Language for

$$r = (aa)^*(bb)^*b$$

$$L(r) = (\{a\}\{a\})^*(\{b\}\{b\})^*\{b\}$$

$$= (\{aa\})^*(\{bb\})^*\{b\}$$

$$= \{aa\}^* \{bb\}^*\{b\}$$

$$= \left\{ a^{2n}b^{2m+1} \mid n \geq 0, m \geq 0 \right\}$$

Example 3 RegExp for

$\{ w \mid w$ consists of alternating 0's and 1's $\}$

Partition

010101…0101	$\longrightarrow$	$(01)^*$
101010…1010	$\longrightarrow$	$(10)^*$
0101010…1010	$\longrightarrow$	$0(10)^*$ or $(01)^*0$
101010…10101	$\longrightarrow$	$(10)^*1$ or $1(01)^*$

The regular expression

$$(01)^*+(10)^*+0(10)^*+(10)^*1$$

Example 3 RegExp for

$\{ w \mid w$ consists of alternating 0's and 1's $\}$

$$\{\varepsilon, 1\}\{010101\ldots0101\}\{\varepsilon, 0\}$$

$$\Downarrow$$

$$\{\varepsilon, 1\}\{0\,1\}^*\{\varepsilon, 0\}$$

$$(\varepsilon + 1)(0\,1)^*(\varepsilon + 0)$$

$$(01)^*+(10)^*+0(10)^*+(10)^*1 \Leftrightarrow (\varepsilon + 1)(0\,1)^*(\varepsilon + 0)$$

$$\Leftrightarrow (\varepsilon + 0)(1\,0)^*(\varepsilon + 1)$$

Example 4 RegExp for

$L=\{w \mid w \in \{0, 1\}^*$ and w has no pair of consecutive 0's $\}$

Partition

no 0	$\{1\}^*$	$\longrightarrow$	1^*
one 0	$\{1\}^*\{0\}\{1\}^*$	$\longrightarrow$	1^*01^*
more 0's	$\{1\}^*\{\{0\}\{1\}\{1\}^*\}^*\{0, \varepsilon\}$		$\{0110111\ldots\ldots\}$

$$1^*(0\,1\,1^*)^*(0+\varepsilon)$$

Example 4 RegExp for

$L=\{w \mid w \in \{0, 1\}^*$ and w has no pair of consecutive 0's $\}$

$\{01\}^*$ $\{\varepsilon, 01, 0101, 010101, 01010101, \ldots\ldots\}$

$\{1, 01\}^*$ $\{\varepsilon, 1, 01, 11, 101, 011, 0101, 111, 1101, 1011, \ldots\ldots\}$

$(1+01)^*$ $\Rightarrow (1+01)^*(0 + \varepsilon)$

Exersizes RegExp for (Σ={0,1})

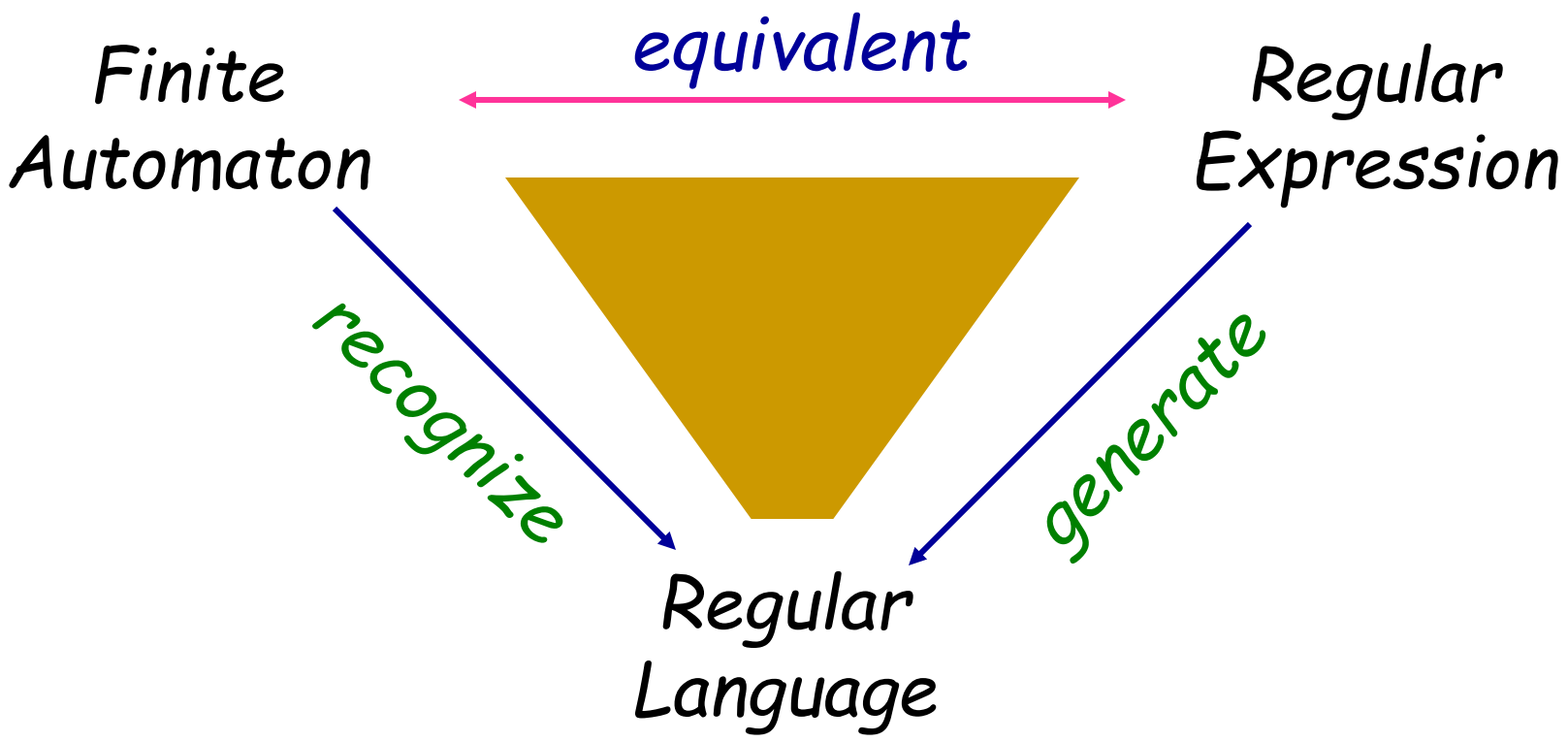
1. {*w* | *w* has exactly a single 1 }

2. {*w* | *w* contains 001 }

3. {*w* | length(w) $\geq$ 3 and the third symbol is 0 }

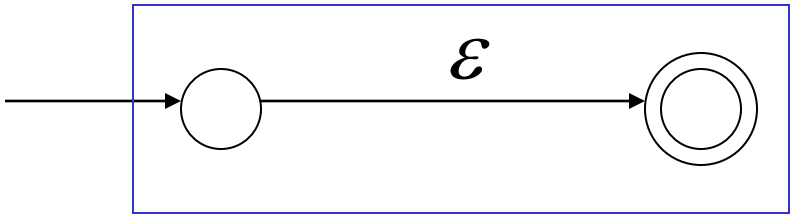
4. What language does the RegEsp ϕ^* represent ?

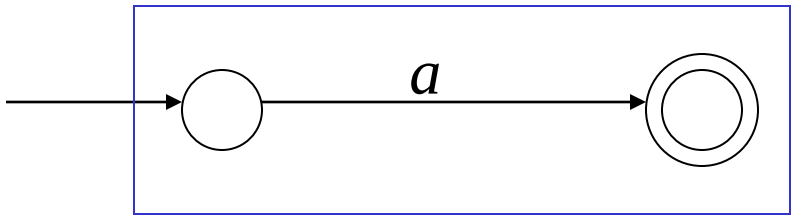
FA & RegExp


Finite Automaton
equivalent
Regular Expression
recognize
generate
Regular Language

RegExp $\Rightarrow$ FA

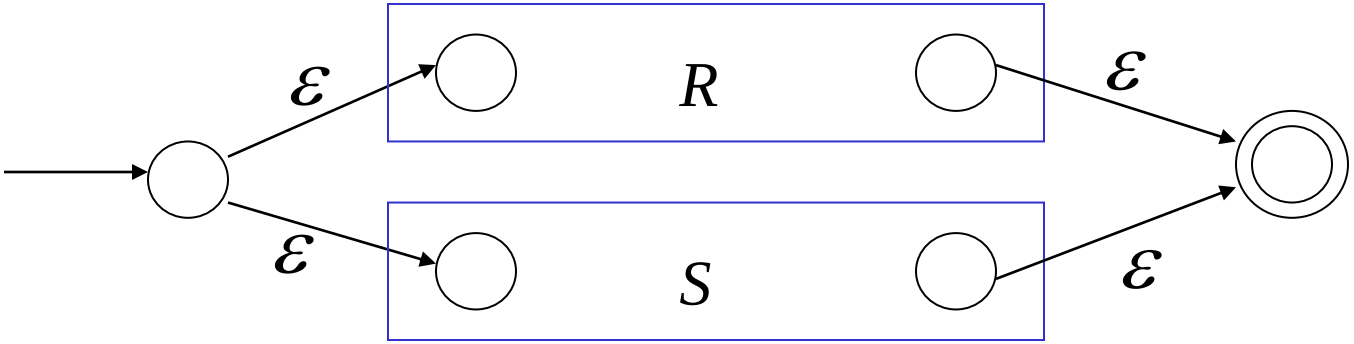
Basis :

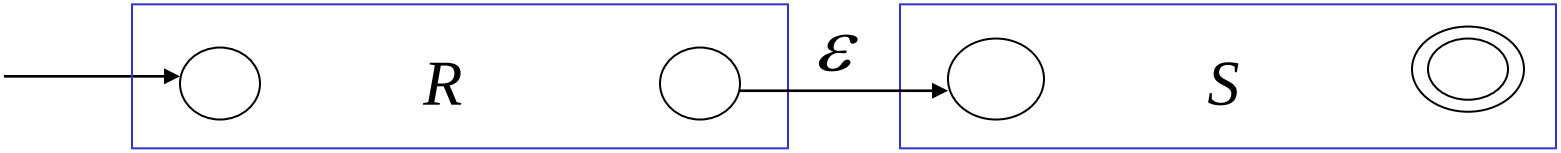


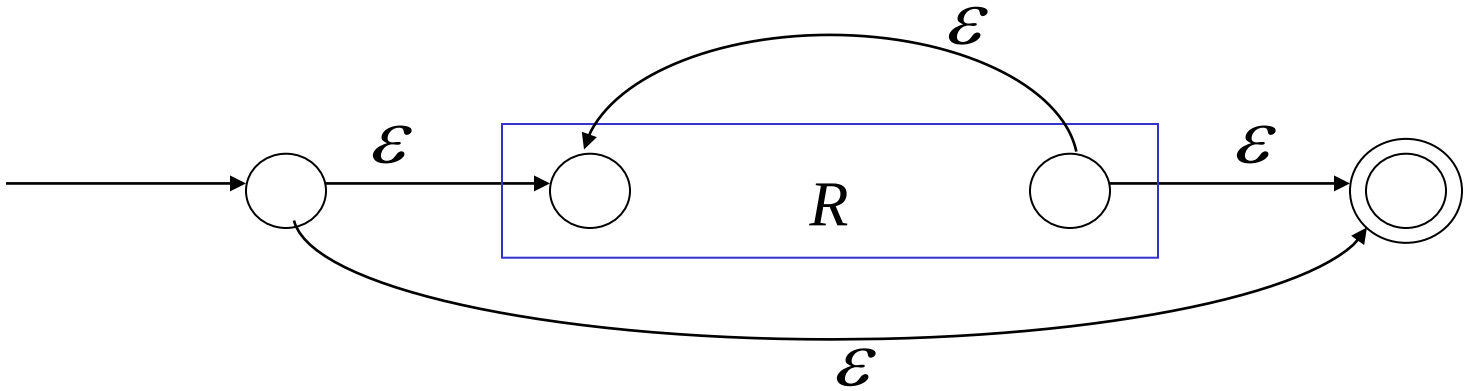


RegExp $\Rightarrow$ FA

Induction :



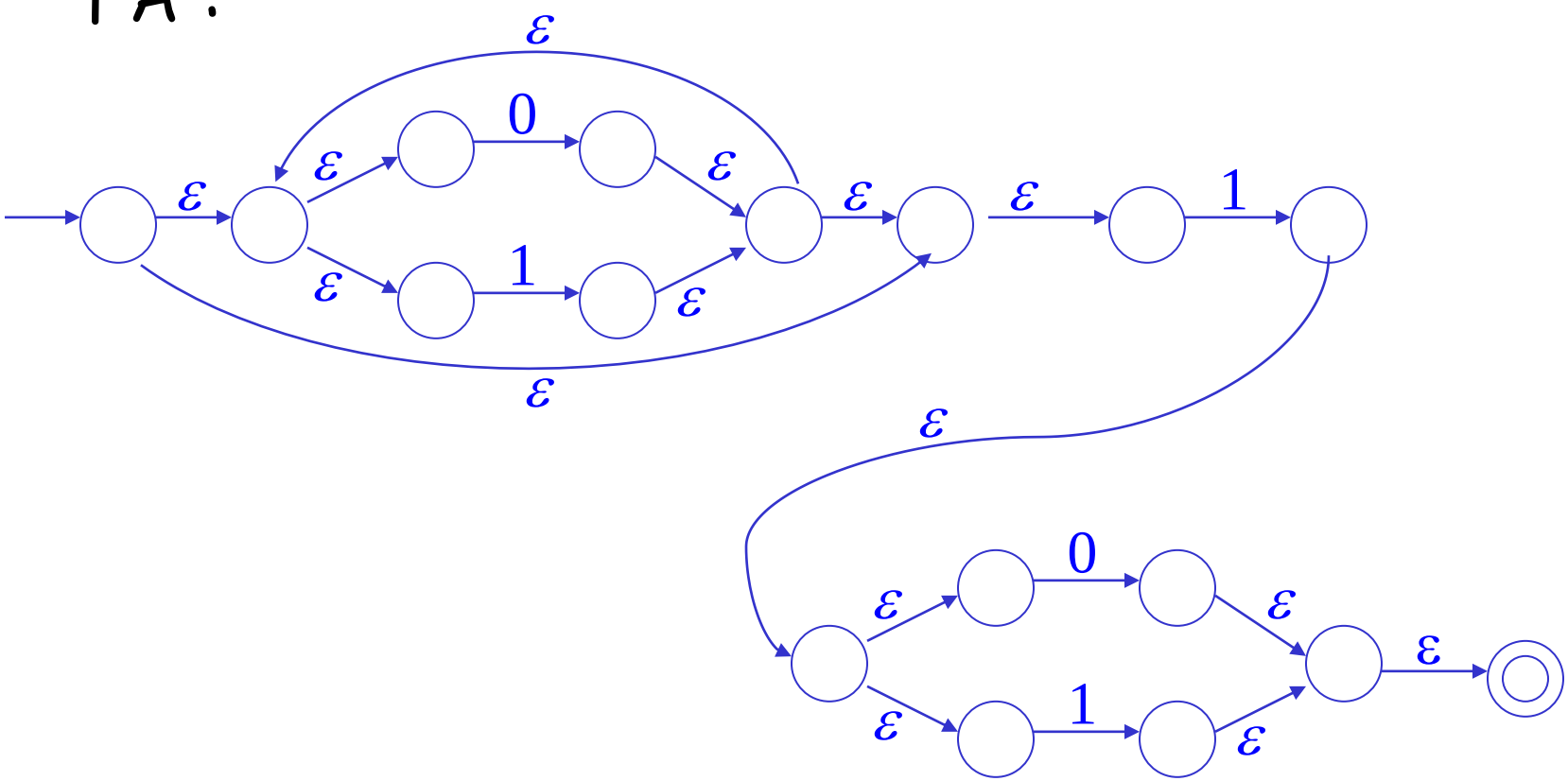




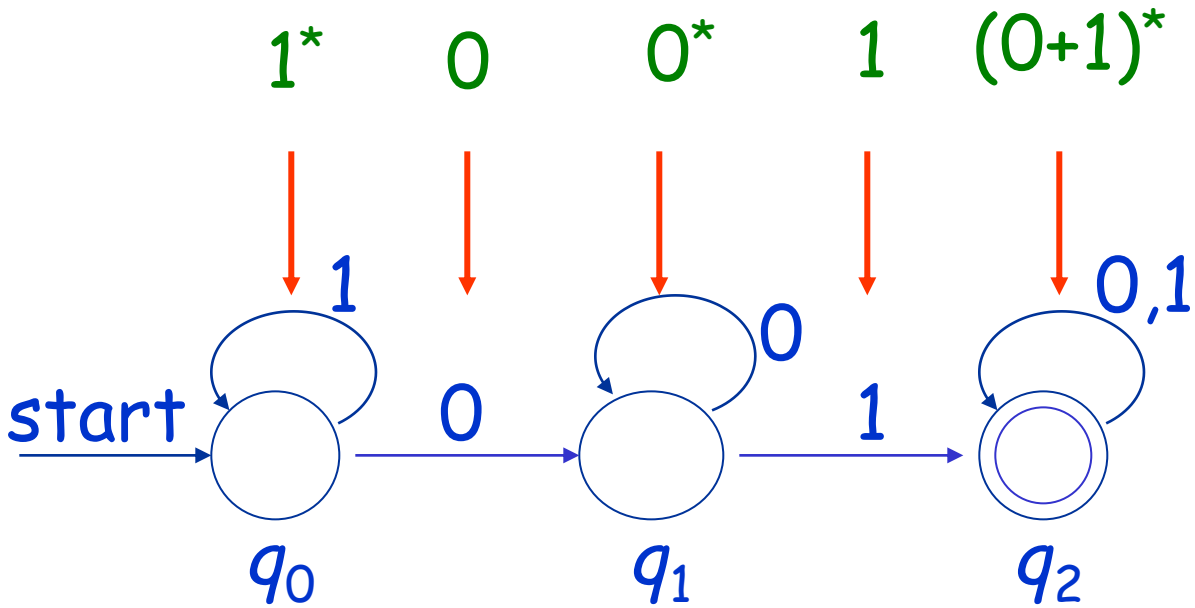
Example 5 RegExp $\Rightarrow$ FA

RegExp : $(0+1)^*1(0+1)$

FA :



Example 6 FA $\Rightarrow$ RegExp

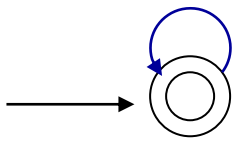


$L=\{w \mid w \in \{0, 1\}^*$ and w contains 01 $\}$

RE : $(0+1)^*01(0+1)^* \Rightarrow 1^*00^*1(0+1)^*$

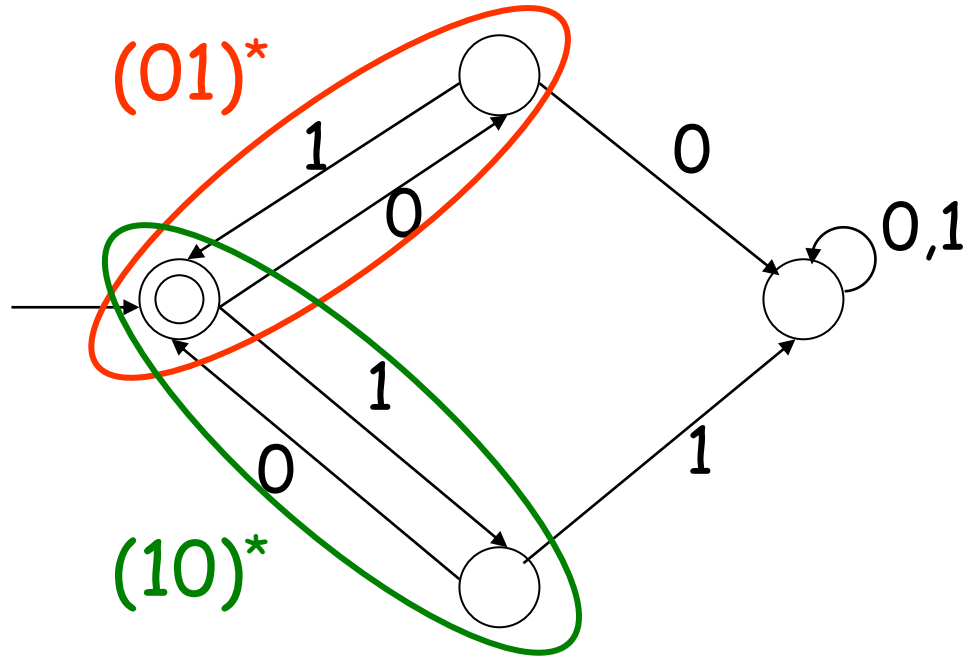
Example 7 FA ⇒ RegExp

01+10



(just notation)

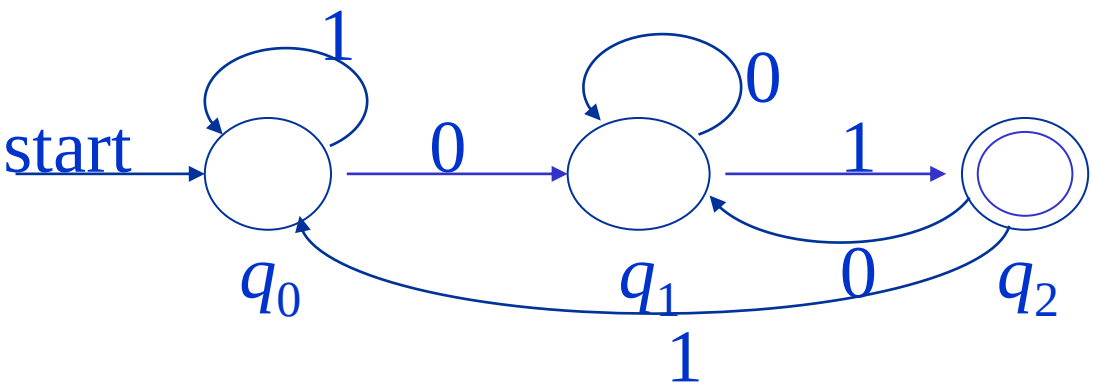
⇐



(01+10)*

Example 8 FA $\Rightarrow$ RegExp

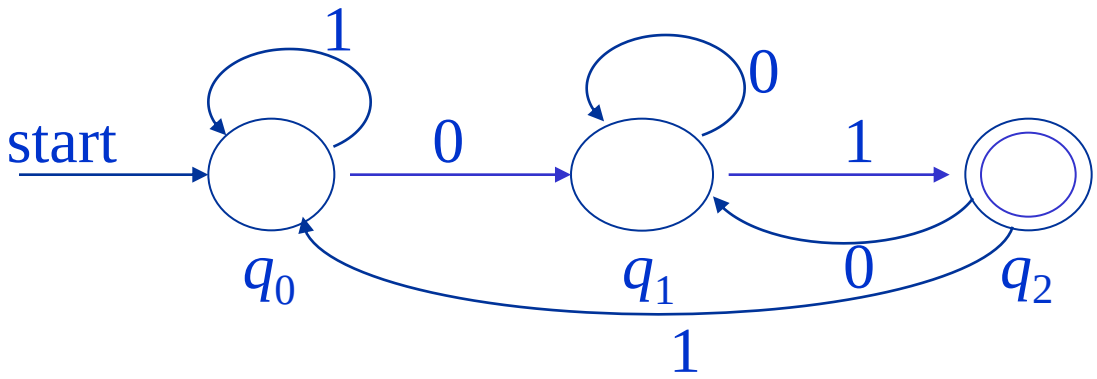
1^* 0 0^* 1 $(00^*1)^*$



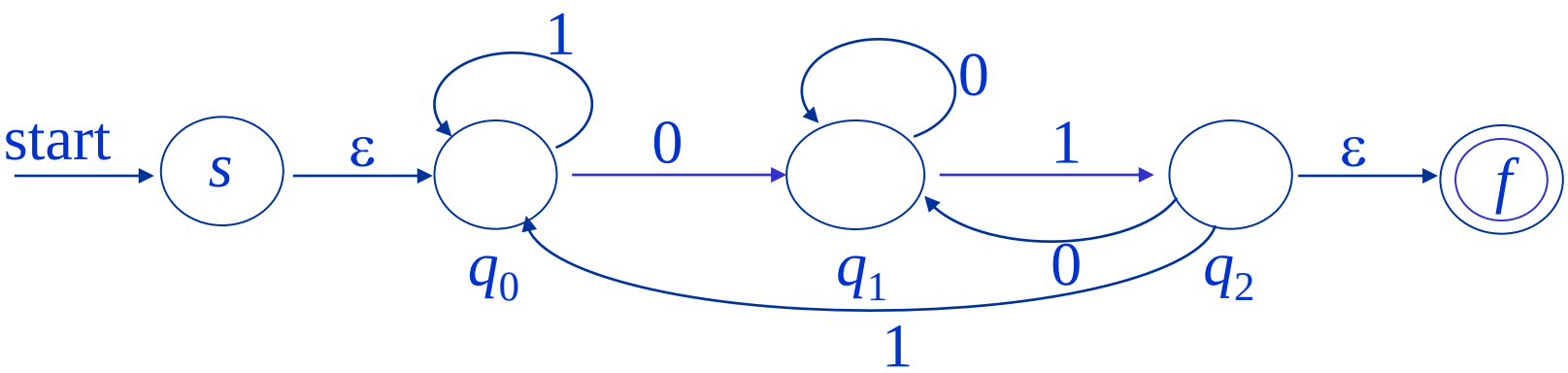
1^* 0 0^* 1 $(11^*00^*1)^*$

$1^*00^*1\ (00^*1+11^*00^*1)^*$

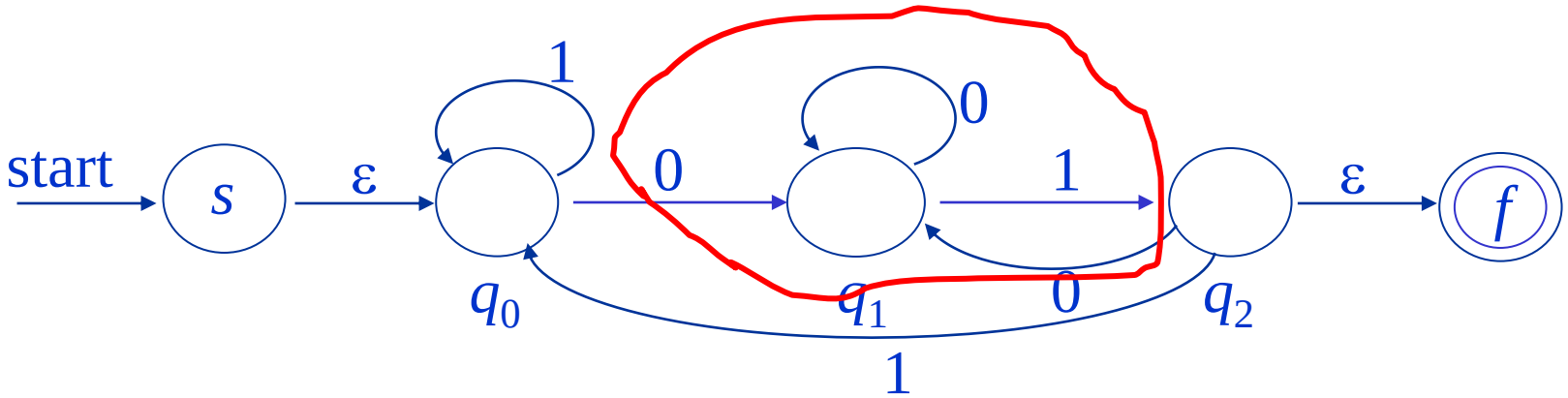
FA ⇒ RegExp – Delete states



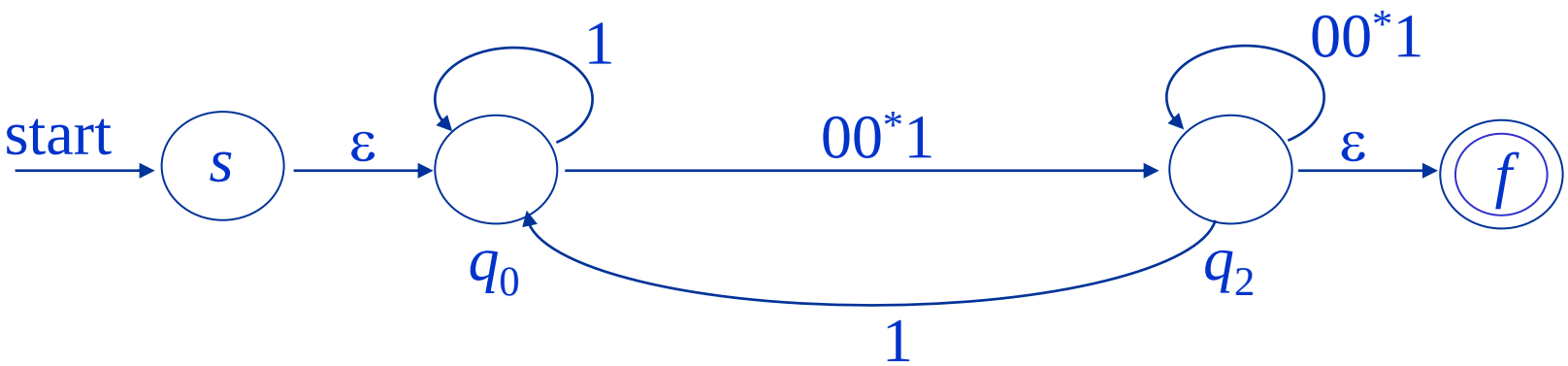
add two states, s and f :



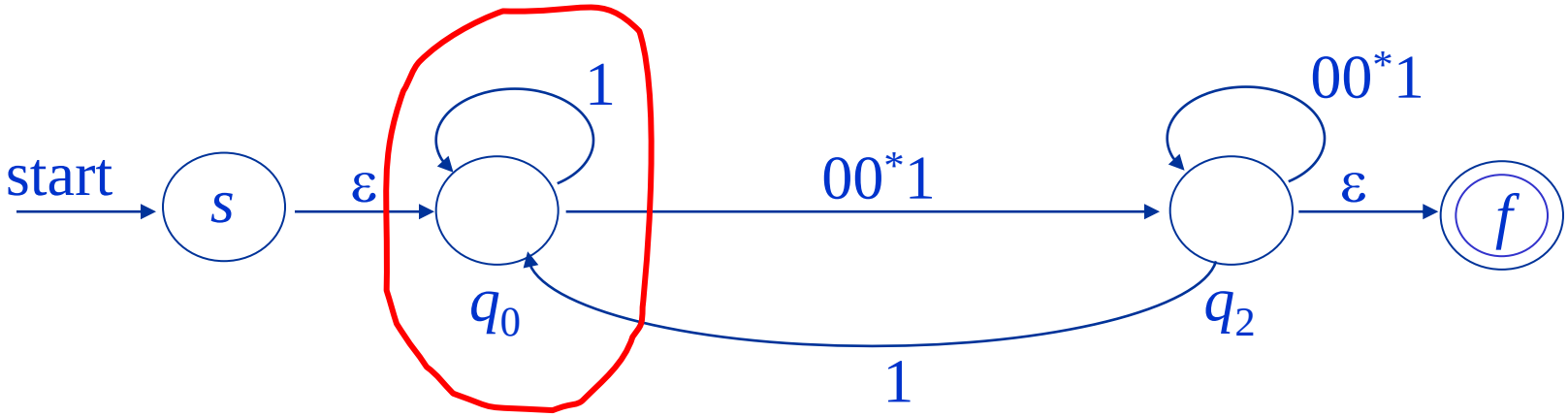
FA ⇒ RegExp – Delete states



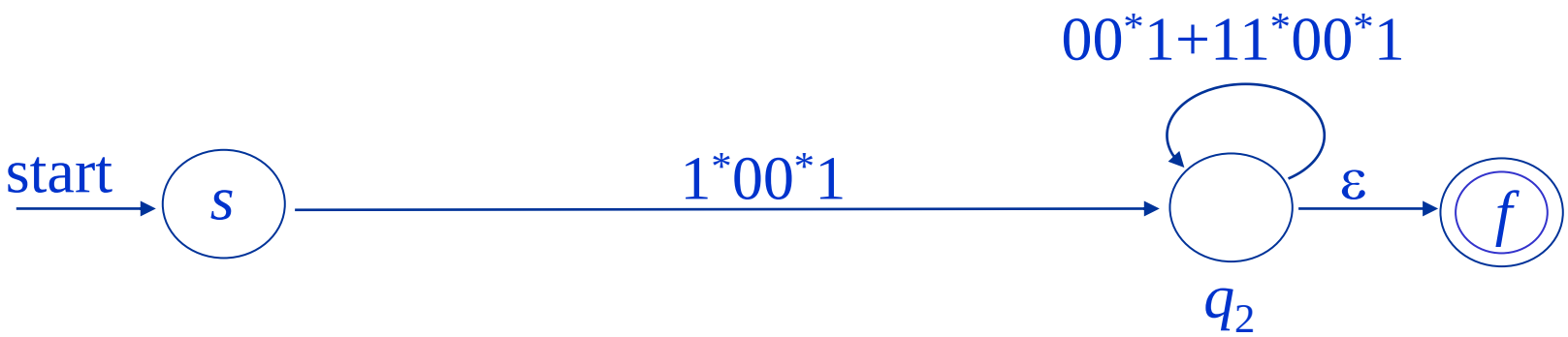
delete q_1 :



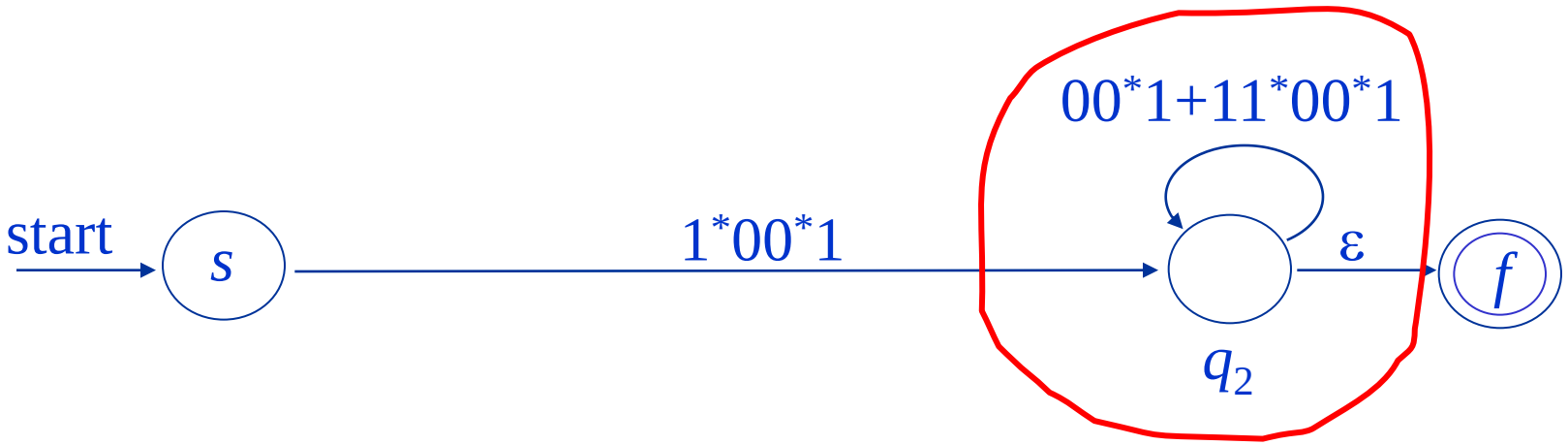
FA ⇒ RegExp – Delete states



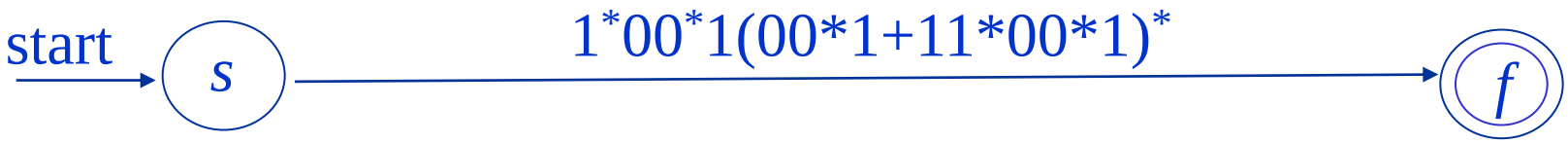
delete q_0 :



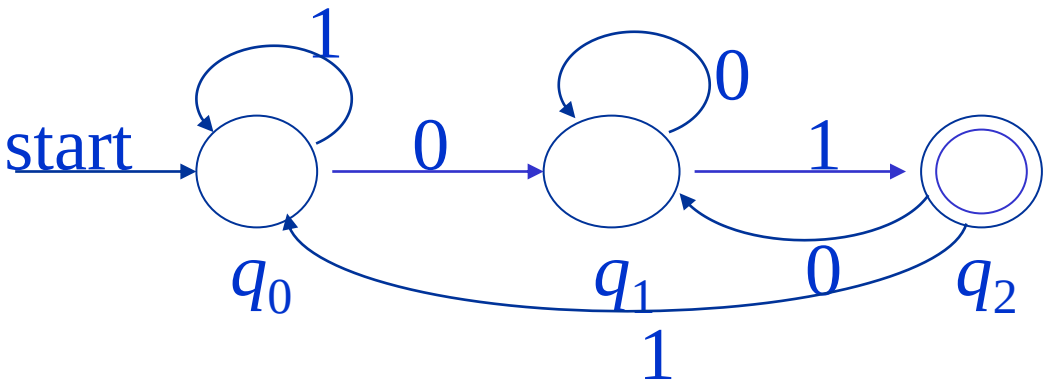
FA ⇒ RegExp – Delete states



delete q_2 :



FA ⇒ RegExp - Induction



- Pick every label on the path from q_0 to q_2

---- one by one

- Form every regexp on the path from q_0 to q_2

---- one by one

FA ⇒ RegExp – Induction

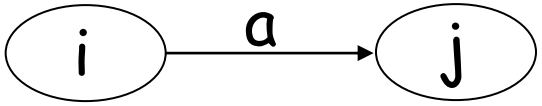
- Q={1,2,3,……,n}
- $R_{ij}^{(k)} : 0 \le k \le n$
  - regular expression of path from *i* to *j*
  - no inner node is greater than *k*

$$R_{ij}^{(k)} \Rightarrow w$$

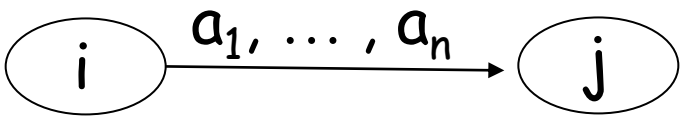
FA $\Rightarrow$ RegExp - Induction

Basis $k = 0,\ i \neq j$

$$\Rightarrow \quad R_{ij}^{(0)} = \phi$$



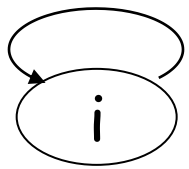
$$\Rightarrow \quad R_{ij}^{(0)} = a$$



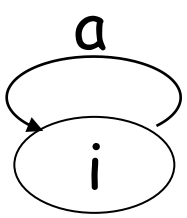
$$\Rightarrow \quad R_{ij}^{(0)} = a_1 + a_2 + \cdots + a_n$$

FA $\Rightarrow$ RegExp - Induction

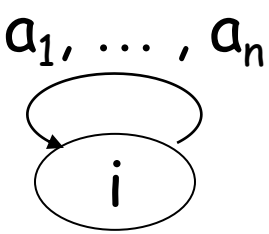
Basis $k = 0$, $i = j$



$$\Rightarrow \quad R_{ij}^{(0)} = \varepsilon + \phi$$



$$\Rightarrow \quad R_{ij}^{(0)} = \varepsilon + a$$

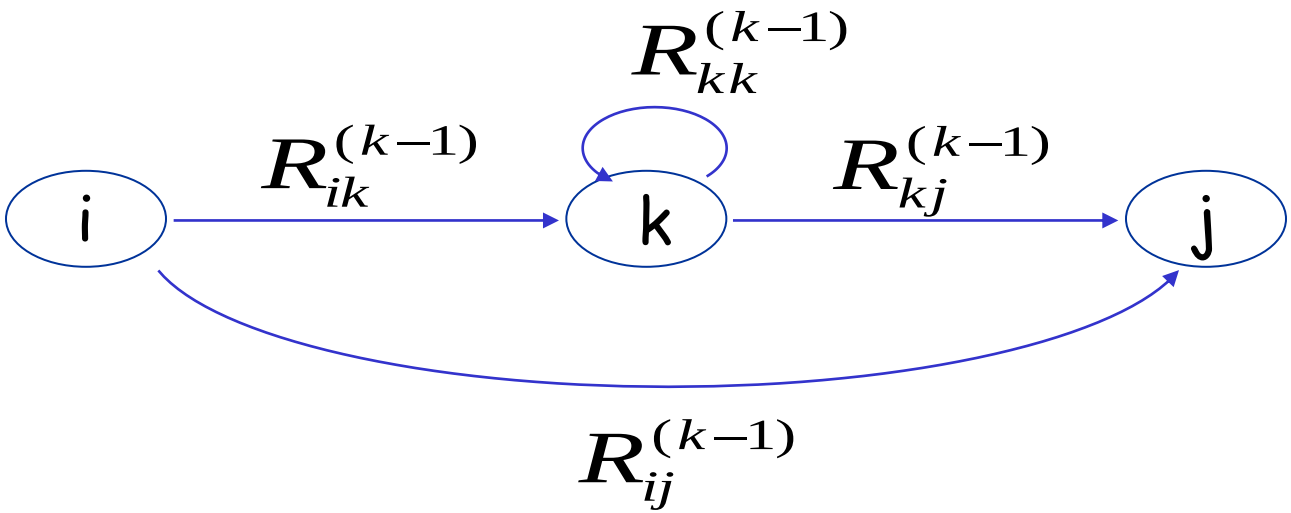


$$\Rightarrow \quad R_{ij}^{(0)} = \varepsilon + a_1 + a_2 + \cdots + a_n$$

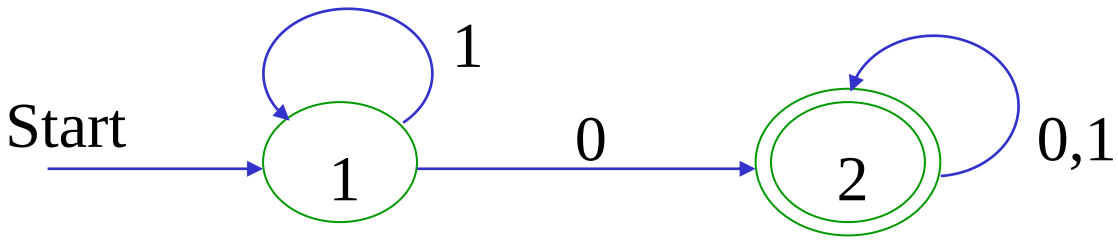
FA $\Rightarrow$ RegExp - Induction

Induction $k \geq 1$

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} (R_{kk}^{(k-1)})^* R_{kj}^{(k-1)}$$



Example 9 FA $\Rightarrow$ RegExp



$$R_{11}^{(0)} = \varepsilon + 1, \quad R_{12}^{(0)} = 0, \quad R_{21}^{(0)} = \phi, \quad R_{22}^{(0)} = \varepsilon + 0 + 1$$

Example 9 FA $\Rightarrow$ RegExp

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} (R_{kk}^{(k-1)})^* R_{kj}^{(k-1)}$$

$$R_{11}^{(0)} = \varepsilon + 1, \quad R_{12}^{(0)} = 0, \quad R_{21}^{(0)} = \phi, \quad R_{22}^{(0)} = \varepsilon + 0 + 1$$

$$R_{11}^{(1)} = \varepsilon + 1 + (\varepsilon + 1)(\varepsilon + 1)^*(\varepsilon + 1) = 1^*$$

$$R_{12}^{(1)} = 0 + (\varepsilon + 1)(\varepsilon + 1)^* 0 = 1^* 0$$

$$R_{21}^{(1)} = \phi + \phi = \phi$$

$$R_{22}^{(1)} = \varepsilon + 0 + 1 + \phi = \varepsilon + 0 + 1$$

Example 9 FA $\Rightarrow$ RegExp

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} (R_{kk}^{(k-1)})^* R_{kj}^{(k-1)}$$

$$R_{11}^{(1)} = 1^*, \; R_{12}^{(1)} = 1^*0, \; R_{21}^{(1)} = \phi, \; R_{22}^{(1)} = \varepsilon + 0 + 1$$

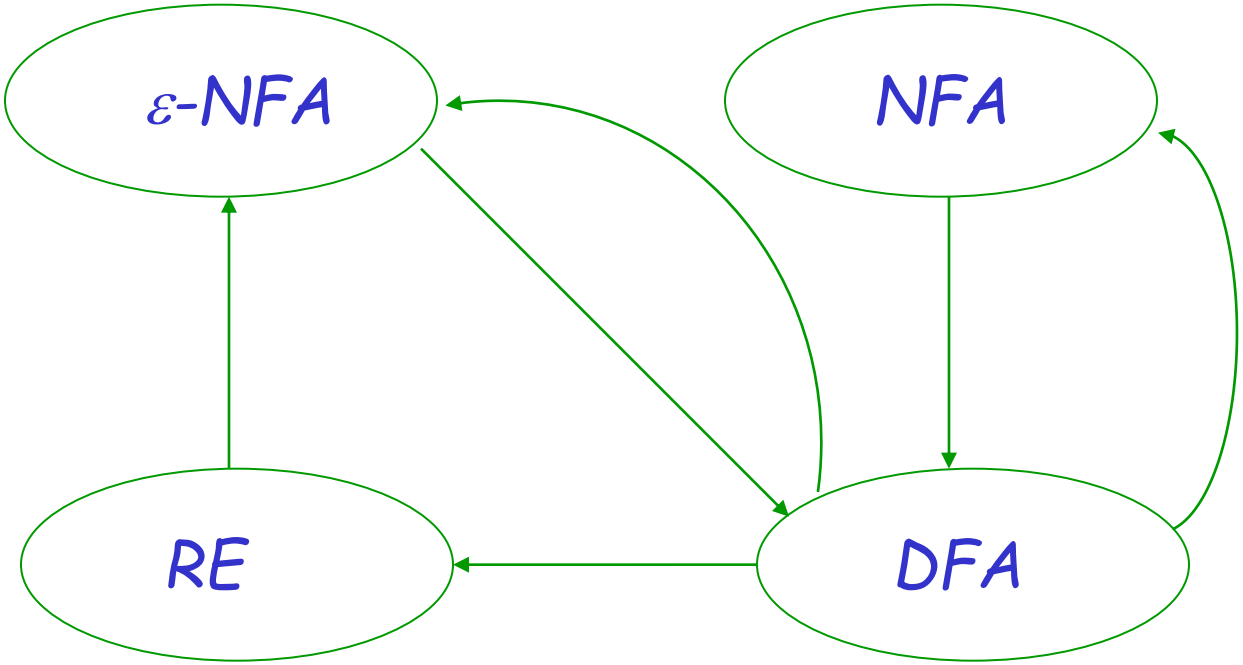
$$R_{11}^{(2)} = 1^* + 1^*0(0+1)^*\phi = 1^*$$

$$R_{12}^{(2)} = 1^*0 + 1^*0(\varepsilon + 0 + 1)^*(\varepsilon + 0 + 1) = 1^*0(0+1)^*$$

$$R_{21}^{(2)} = \phi + \phi = \phi$$

$$R_{22}^{(2)} = \varepsilon + 0 + 1 + (\varepsilon + 0 + 1)^* = (0+1)^*$$

FA & RE



Good good study day day up !